Apply filters to SQL queries

Project description

In this project, I used SQL queries to investigate potential security issues within an organization's database. I analyzed login attempts to identify suspicious activity, such as failed logins outside business hours and login attempts from specific countries and dates. I also filtered employee records to identify specific departments and office locations that required security updates.

Retrieve after hours failed login attempts

```
SELECT *
FROM log_in_attempts
WHERE login_time > '18:00' AND success = FALSE;
```

Explanation: This query retrieves all login attempts that occurred after 6:00 PM and were unsuccessful. The `WHERE` clause filters the records using two conditions. The `AND` operator ensures that both conditions must be true: the login time must be later than 18:00, and the success column must contain FALSE, indicating a failed login attempt.

Retrieve login attempts on specific dates

```
SELECT *
FROM log_in_attempts
WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

Explanation: This query retrieves all login attempts that occurred on May 9, 2022, or May 8, 2022. The `WHERE` clause filters the records based on the login_date column. The `OR` operator allows records matching either date to be included in the results.

Retrieve login attempts outside of Mexico

```
SELECT *
FROM log_in_attempts
WHERE NOT country LIKE '%MEX%';
```

Explanation: This query retrieves all login attempts that occurred outside of Mexico. The LIKE operator combined with the wildcard character % searches for country values containing the text "MEX", including both "MEX" and "MEXICO". The NOT operator excludes all records matching this pattern, returning only login attempts from other countries.

Retrieve employees in Marketing

```
SELECT *
FROM employees
WHERE department = 'Marketing' AND office LIKE 'East%';
```

Explanation: This query retrieves employees who belong to the Marketing department and work in any office located in the East building. The `AND` operator ensures that both conditions must be met. The `LIKE` operator and the wildcard character `%` match all office values that begin with "East", such as East-170 and East-320.

Retrieve employees in Finance or Sales

```
SELECT *
FROM employees
WHERE department = 'Finance' OR department = 'Sales';
```

Explanation:This query retrieves all employees who belong to either the Finance or Sales department. The `OR` operator allows records to match either of the two conditions, so employees from both departments are included in the results.

Retrieve all employees not in IT

```
SELECT *
FROM employees
WHERE NOT department = 'Information Technology';
```

This query retrieves all employees who do not belong to the Information Technology department. The `NOT` operator excludes employees whose department is Information Technology, returning employees from all other departments.

Summary

In this project, I used SQL filtering techniques to investigate potential security incidents and retrieve relevant employee information. I practiced using `AND`, `OR`, and `NOT` operators to combine and exclude conditions, as well as the `LIKE` operator and wildcard characters to search for specific patterns. I also applied date and time filters to identify suspicious login attempts. These techniques are useful for security investigations, monitoring login activity, and identifying systems and employees that require security updates.